

Quiz 10: Random Walks, Convergence, and Correlation

Throughout problems 1–3: Let $(X_n)_{n \geq 0}$ be a simple symmetric random walk on $\mathbb{Z}$,

$$X_0 = 0, \quad X_n = \xi_1 + \cdots + \xi_n,$$

where the $\xi_1, \xi_2, \dots$ are i.i.d. with

$$\mathbb{P}(\xi_1 = 1) = \mathbb{P}(\xi_1 = -1) = \frac{1}{2}.$$

1. (2 points) **PMF of a simple random walk.**

Show that for any integer k ,

$$\mathbb{P}(X_n = k) = \begin{cases} \binom{n}{(n+k)/2} \frac{1}{2^n}, & \text{if } |k| \leq n \text{ and } k \text{ has the same parity as } n, \\ 0, & \text{otherwise.} \end{cases}$$

2. (2 points) **Covariance and correlation.**

Fix integers $1 \leq m \leq n$.

- (a) Show that $\text{Cov}(X_m, X_n) = m$.
- (b) Deduce that $\text{Corr}(X_m, X_n) = \sqrt{\frac{m}{n}}$.
- (c) Interpret this formula when $m \ll n$ and when $m \approx n$.

3. (2 points) **Normal scaling of a random walk.**

Define

$$Z_n := \frac{X_n}{\sqrt{n}}.$$

- (a) Compute $\mathbb{E}[Z_n]$ and $\text{Var}(Z_n)$.
- (b) For $t > 0$, show that $\mathbb{P}(|Z_n| \geq t) \leq \frac{1}{t^2}$.
- (c) Compare with $\frac{X_n}{n}$ and explain the difference in behavior.

4. (2 points) **A Gaussian random walk with shrinkage.**

Let $-1 < \rho < 1$. Let $Z_0, Z_1, Z_2, \dots$ be independent $\mathcal{N}(0, 1)$ random variables, and define

$$Y_0 = Z_0, \quad Y_{n+1} = \rho Y_n + \sqrt{1 - \rho^2} Z_{n+1}.$$

- (a) Show $\mathbb{E}[Y_n] = 0$ and $\text{Var}(Y_n) = 1$.
- (b) Show $\text{Cov}(Y_m, Y_n) = \rho^{n-m}$ for $m \leq n$.
- (c) Describe how dependence changes as $n - m$ grows.